

Executive Data Warehouse Architecture Report: Airline Operations & Analytics

1. Data Warehouse Models Detail

The Data Warehouse (DWH) utilizes a **Galaxy Schema** (Fact Constellation) approach. This architecture deploys multiple fact tables that share common, conformed dimensions to accommodate the differing grains of airline business processes without creating sparse or overly complex single tables.

Dimension Models

Dimensions provide the descriptive context for the fact measurements.

- **DIM_DATE (Role-Playing Dimension):** Generated sequentially to handle all time-based reporting (Booking Date, Flight Date, Interaction Date). Includes calendar hierarchies (Year, Quarter, Month, Day) and weekend flags.
- **DIM_PASSENGER (Slowly Changing Dimension - Type 2):** Tracks the lifecycle of a customer. Because frequent flyers upgrade or downgrade their tiers, this table utilizes `EFFECTIVE_DATE`, `EXPIRATION_DATE`, and `IS_CURRENT` flags to preserve historical accuracy regarding their status at the time of a specific flight.
- **DIM_FLIGHT:** A flattened, denormalized model combining original OLTP routes, airports, and aircraft data into a single table for fast lookup of origins, destinations, and distances.
- **DIM_FARE_CLASS, DIM_CHANNEL, DIM_PROMOTION, DIM_INTERACTION (Standard Dimensions):** Lookup tables containing relatively static descriptive attributes mapping to tickets, booking mediums, marketing campaigns, and customer support ticket categories.

Fact Models

Fact tables contain the quantitative metrics for specific business processes.

- **FACT_FLIGHT_ACTIVITY (Transaction Fact):** Grain is set at one row per passenger per completed flight segment. It consolidates flight activity, miles accrued/redeemed, upgrades, and promotional responses to serve Marketing Department KPIs.
- **FACT_RESERVATION_FINANCES (Transaction Fact):** Grain is set at one row per ticket issued. It centralizes all revenue and cost metrics (`FINAL_TICKET_AMOUNT_USD`, `EST_FLIGHT_COST_ALLOCATION`, `PROFIT_MARGIN_USD`) to serve the Finance Department.
- **FACT_CUSTOMER_CARE (Accumulating Snapshot Fact):** Grain is set at one row per customer interaction. It tracks the lifecycle of an inquiry or complaint, from initiation to resolution, capturing metrics like resolution time and customer satisfaction (`CSAT`) scores.

Data Mart Models

- **MART_CHANNEL_PROFITABILITY:** Aggregates finance data at the Channel and Month level to provide immediate dashboarding for revenue and profit margins.
- **MART_FREQUENT_FLYER_BEHAVIOR:** Aggregates marketing data at the Passenger and Month level to track overall loyalty and engagement.

2. Indexing and Partitioning Strategy

Partitioning Strategy

Range Partitioning is applied to all fact tables (`FACT_FLIGHT_ACTIVITY`, `FACT_RESERVATION_FINANCES`, `FACT_CUSTOMER_CARE`) based on the `DATE_KEY`.

- **Why:** Fact tables in an airline environment grow massively. Partitioning by year/month significantly improves query performance (Partition Pruning) since executives typically query specific timeframes (e.g., "Q3 2026"). It also allows for easier data archiving and maintenance operations.

Indexing Strategy

- **Bitmap Indexes:** Applied to low-cardinality foreign keys within the Fact tables (e.g., `FARE_CLASS_SK`, `CHANNEL_SK`, `PROMOTION_SK`, `INTERACTION_SK`).
 - **Why:** Bitmap indexes are highly optimized for Data Warehouses where columns have a small number of distinct values (e.g., only 4-5 booking channels). They allow rapid aggregation and filtering.
- **B-Tree Indexes:** Applied to high-cardinality foreign keys (e.g., `PASSENGER_SK`, `FLIGHT_SK`).
 - **Why:** Standard B-Tree indexes provide efficient lookups and joins for columns containing millions of unique identifiers, such as passenger or flight surrogate keys.

3. Business Questions, SQL Queries, and Outputs

Note: The outputs provided below are generated against a representative analytical dataset demonstrating the logical design.

Q1: What proportion of frequent flyers have Gold, Platinum, or Titanium status?

1	--Query 1: What proportion of frequent flyers have Gold, Platinum, or Titanium status?
2	SELECT
3	TIER_NAME,
4	COUNT(PASSENGER_SK) AS PASSENGER_COUNT,
5	ROUND(COUNT(PASSENGER_SK) * 100.0 / SUM(COUNT(PASSENGER_SK)) OVER (), 2) AS PROPORTION_PCT
6	FROM DIM_PASSENGER
7	WHERE IS_CURRENT = 'A'
8	AND TIER_NAME IN ('Gold', 'Platinum', 'Titanium')
9	GROUP BY TIER_NAME;

TIER_NAME	PASSENGER_COUNT	PROPORTION_PCT
Gold	35	49.3
Titanium	11	15.49
Platinum	25	35.21

Q2: How often do frequent flyers upgrade?

10	
11	-- Query 2: How often do frequent flyers upgrade?
12	SELECT
13	p.TIER_NAME,
14	COUNT(f.FLIGHT_ACTIVITY_ID) AS TOTAL_FLIGHTS,
15	SUM(f.IS_UPGRADED) AS TOTAL_UPGRADES,
16	ROUND(SUM(f.IS_UPGRADED) * 100.0 / COUNT(f.FLIGHT_ACTIVITY_ID), 2) AS UPGRADE_RATE_PCT
17	FROM FACT_FLIGHT_ACTIVITY f
18	JOIN DIM_PASSENGER p ON f.PASSENGER_SK = p.PASSENGER_SK
19	WHERE p.TIER_NAME != 'None' -- Assuming 'None' is for non-loyalty members
20	GROUP BY p.TIER_NAME;
21	

TIER_NAME	TOTAL_FLIGHTS	TOTAL_UPGRADES	UPGRADE_RATE_PCT
Silver	36	13	36.11
Gold	37	11	29.73
Blue	26	7	26.92
Titanium	26	9	34.62
Platinum	35	8	22.86

Q3: Do frequent flyers respond to special fare promotions?

22	--Query 3: Do frequent flyers respond to special fare promotions?
23	SELECT
24	p.TIER_NAME,
25	pr.PROMO_NAME,
26	COUNT(f.FLIGHT_ACTIVITY_ID) AS TOTAL_BOOKINGS,
27	SUM(f.RESPONDED_TO_PROMO) AS PROMO_RESPONSES,
28	ROUND(SUM(f.RESPONDED_TO_PROMO) * 100.0 / COUNT(f.FLIGHT_ACTIVITY_ID), 2) AS RESPONSE_RATE_PCT
29	FROM FACT_FLIGHT_ACTIVITY f
30	JOIN DIM_PASSENGER p ON f.PASSENGER_SK = p.PASSENGER_SK
31	JOIN DIM_PROMOTION pr ON f.PROMOTION_SK = pr.PROMOTION_SK
32	WHERE pr.PROMO_ID != -1 -- Excluding flights without promotions
33	GROUP BY p.TIER_NAME, pr.PROMO_NAME;

Data Grid	Auto Trace	DBMS Output (disabled)	Query Viewer	CodeXpert	Explain Plan	Script Output
Cancel						
TIER_NAME	PROMO_NAME	TOTAL_BOOKINGS	PROMO_RESPONSES	RESPONSE_RATE_PCT		
Gold	Summer Sale 2024	3	0	0		
Platinum	Cairo-Dubai Special	6	0	0		
Gold	New Member Discount	2	0	0		
Blue	Eid Al-Fitr Offer	10	0	0		
Blue	Winter Deals 2024	1	1	100		
Platinum	Business Miles Boost	3	0	0		
Gold	Business Miles Boost	3	0	0		
Silver	Eid Al-Fitr Offer	15	2	13.33		
Silver	New Member Discount	1	0	0		
Platinum	Ramadan Special	6	0	0		
Gold	Winter Deals 2024	1	0	0		
Titanium	Business Miles Boost	6	2	33.33		
Blue	Summer Sale 2024	6	1	16.67		
Titanium	Ramadan Special	7	0	0		
Gold	Ramadan Special	7	1	14.29		

Q4: How long are their overnight stays?

```

39      --Query 4: How long are their overnight stays?
40      SELECT
41          p.TIER_NAME,
42          AVG(f.OVERNIGHT_STAY_NIGHTS) AS AVG_STAY_DURATION_NIGHTS,
43          MAX(f.OVERNIGHT_STAY_NIGHTS) AS MAX_STAY_DURATION_NIGHTS
44      FROM FACT_FLIGHT_ACTIVITY f
45      JOIN DIM_PASSENGER p ON f.PASSENGER_SK = p.PASSENGER_SK
46      WHERE f.OVERNIGHT_STAY_NIGHTS > 0
47      GROUP BY p.TIER_NAME;
48

```

Data Grid			
	Data Grid	Auto Trace	DBMS Output (disabled)
	Query Viewer	CodeXpert	Exp
	Cancel		
TIER_NAME	AVG_STAY_DURATION_NIGHTS	MAX_STAY_DURATION_NIGHTS	
Silver	3	7	
Gold	2.57142857142857	7	
Blue	2.68421052631579	5	
Titanium	2.6	5	
Platinum	2.53571428571429	5	

Q5: Analyze company profit through multiple reservation channels

49 --Query 5: Analyze company profit through multiple reservation channels

50 ▶ SELECT

51 c.CHANNEL_NAME,

52 c.CHANNEL_TYPE,

53 SUM(f.FINAL_TICKET_AMOUNT_USD) AS TOTAL_REVENUE,

54 SUM(f.EST_FLIGHT_COST_ALLOCATION) AS TOTAL_COST,

55 SUM(f.PROFIT_MARGIN_USD) AS TOTAL_PROFIT

56 FROM FACT_RESERVATION_FINANCES f

57 JOIN DIM_CHANNEL c ON f.CHANNEL_SK = c.CHANNEL_SK

58 GROUP BY c.CHANNEL_NAME, c.CHANNEL_TYPE

59 ORDER BY TOTAL_PROFIT DESC;

60

Data Grid

Data Grid | Auto Trace | DBMS Output (disabled) | Query Viewer | CodeXpert | Explain Plan | Script Output

Cancel

CHANNEL_NAME	CHANNEL_TYPE	TOTAL_REVENUE	TOTAL_COST	TOTAL_PROFIT
▶ Travel Agent	Human	59696	3786.68	55909.32
Website	Digital	37783	5270.6	32512.4
Call Center	Human	24125	3561.76	20563.24
Mobile App	Digital	17822	3769.29	14052.71
Airport Kiosk	Digital	2221	765.44	1455.56

Q6: Analyze customer care interaction type and problem severity

61 --Query 6: Analyze customer care interaction type and problem severity

62 ▶ SELECT

63 i.INTERACTION_TYPE,

64 i.SEVERITY,

65 COUNT(c.CUSTOMER_CARE_ID) AS TOTAL_ISSUES,

66 AVG(c.RESOLUTION_TIME_DAYS) AS AVG_RESOLUTION_DAYS,

67 AVG(c.CSAT_SCORE) AS AVG_CUSTOMER_SATISFACTION

68 FROM FACT_CUSTOMER_CARE c

69 JOIN DIM_INTERACTION i ON c.INTERACTION_SK = i.INTERACTION_SK

70 GROUP BY i.INTERACTION_TYPE, i.SEVERITY

71 ORDER BY TOTAL_ISSUES DESC;

72

Data Grid

Data Grid | Auto Trace | DBMS Output (disabled) | Query Viewer | CodeXpert | Explain Plan | Script Output

Cancel

INTERACTION_TYPE	SEVERITY	TOTAL_ISSUES	AVG_RESOLUTION_DAYS	AVG_CUSTOMER_SATISFACTION
▶ SERVICE_REQUEST	LOW	5	1.8	3
COMPLAINT	MEDIUM	4	2.5	3.25
COMPLAINT	LOW	4	2.5	3.75
INQUIRY	LOW	2	3.5	3
COMPLAINT	CRITICAL	1	0	4
FEEDBACK	LOW	1	1	2
INQUIRY	MEDIUM	1	5	4

